WORKED SOLUTION

STEP III 2021, Question 2

Pure • Determinants and Inequalities

STEP III 2021, Question 2

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

Each definition of x, y, z can be cleared of fractions to give an equation that is *linear* in a, b, c. Three such equations make a homogeneous linear system with a non-zero solution (a, b, c), so the matrix of coefficients must be singular: its determinant is zero. That determinant is exactly the symmetric expression in x, y, z we want. After that, the inequalities come from rewriting the target in terms of x, y, z and using basic facts about sums of squares and AM–GM.

Solution

Part (i)

Since a, b, c are distinct, none of the denominators is zero. Clearing fractions,

$$x(b-c) = a, \qquad y(c-a) = b, \qquad z(a-b) = c.$$

Now multiply out the matrix product row by row:

$$\begin{aligned} a - xb + xc &= a - x(b-c) = a - a = 0, \\ ya + b - yc &= b - y(c-a) = b - b = 0, \\ -za + zb + c &= c - z(a-b) = c - c = 0, \end{aligned}$$

which is the required matrix equation.

Call the 3×3 matrix M. Because a, b, c are distinct, they are not all zero, so M sends a *non-zero* vector to the zero vector.

Ask yourself Why does this force $\det M = 0$?

If $\det M \neq 0$ then M^{-1} exists, and multiplying $M\mathbf{v} = \mathbf{0}$ on the left by M^{-1} would give

$\mathbf{v} = \mathbf{0}$, contradicting $(a, b, c) \neq (0, 0, 0)$. So $\det M = 0$. Expanding along the first row,

$$\begin{aligned}\det M &= 1 \cdot (1 + yz) + x(y - yz) + x(yz + z) \\ &= 1 + yz + xy - xyz + xyz + xz \\ &= 1 + xy + yz + zx.\end{aligned}$$

Setting this equal to zero gives $yz + zx + xy = -1$.

For the inequality, the left-hand side is $x^2 + y^2 + z^2$. Using the identity $(x + y + z)^2 = x^2 + y^2 + z^2 + 2(xy + yz + zx)$,

$$x^2 + y^2 + z^2 = (x + y + z)^2 - 2(xy + yz + zx) = (x + y + z)^2 + 2 \geq 2,$$

since the square of the real number $x + y + z$ is non-negative. This is the required result.

Part (ii)

Here the denominators are positive, and clearing fractions gives $x(b + c) = 2a$ and so on. Rearranged as linear equations in a, b, c:

$$\begin{pmatrix} -2 & x & x \\ y & -2 & y \\ z & z & -2 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

(The first row reads $-2a + x(b + c) = 0$, and the others are similar.) As $a, b, c > 0$ the vector is non-zero, so, exactly as in part (i), the determinant vanishes:

$$\begin{aligned}0 &= -2(4 - yz) - x(-2y - yz) + x(yz + 2z) \\ &= -8 + 2yz + 2xy + xyz + xyz + 2xz \\ &= -8 + 2(xy + yz + zx) + 2xyz.\end{aligned}$$

Dividing by 2 gives $xyz + yz + zx + xy = 4$.

The first inequality. The trick is to divide each bracket on the left by the matching bracket on the right:

$$\frac{2a + b + c}{b + c} = \frac{2a}{b + c} + 1 = 1 + x,$$

and similarly for the other two. Since $(b + c)(c + a)(a + b) > 0$, the inequality is equivalent to $(1 + x)(1 + y)(1 + z) > 5$. Expanding and using the result just proved,

$$(1 + x)(1 + y)(1 + z) = 1 + (x + y + z) + (xy + yz + zx) + xyz = 5 + (x + y + z).$$

As x, y, z are all positive, this exceeds 5, which proves the first inequality.

The second inequality. By the same calculation we now need $x + y + z > 2$. Write $S = x + y + z$.

Ask yourself We know $xy + yz + zx + xyz = 4$. How can that force S to be reasonably large?

Both $xy + yz + zx$ and xyz are bounded *above* in terms of S:

- $S^2 - 3(xy + yz + zx) = \frac{1}{2}\left[(x-y)^2 + (y-z)^2 + (z-x)^2\right] \geq 0$, so $xy + yz + zx \leq \frac{1}{3}S^2$;
- by AM–GM for the positive numbers x, y, z, we have $xyz \leq \left(\frac{1}{3}S\right)^3 = \frac{1}{27}S^3$.

Adding,

$$4 = xy + yz + zx + xyz \leq \tfrac{1}{3}S^2 + \tfrac{1}{27}S^3.$$

The right-hand side is an increasing function of S for $S > 0$, and it equals $3 + 1 = 4$ when $S = 3$. So if $S < 3$ the right-hand side would be less than 4, which is impossible. Hence $S \geq 3$, and

$$\frac{(2a+b+c)(a+2b+c)(a+b+2c)}{(b+c)(c+a)(a+b)} = 5 + S \geq 8 > 7,$$

which proves the second inequality (and in fact gives the sharper bound 8, attained when $a = b = c$).

Checks

- Take $a = b = c = 1$ in part (ii): the left-hand side is $4^3 = 64$ and $(b+c)(c+a)(a+b) = 8$, so the ratio is 8, matching $5 + S$ with $x = y = z = 1$, $S = 3$. Also $xyz + xy + yz + zx = 1 + 3 = 4$.
- In part (i), take $(a, b, c) = (0, 1, 2)$: then $x = 0$, $y = \frac{1}{2}$, $z = -2$, and $xy + yz + zx = -1$ as required; the sum of squares is $\frac{17}{4} \geq 2$.
- A computer check over 20 000 random rational triples confirmed both identities exactly; the smallest ratio found in part (ii) was about 8.0001, and for $a = b = 1$, $c = 10^{-6}$ it is about 9.0, so the bound 8 is the true minimum, comfortably above 7.

Common slips

- Asserting $\det M = 0$ without saying why: the key point is that (a, b, c) is a *non-zero* vector, which needs "distinct" in part (i) and "positive" in part (ii).
- Dividing $2a + b + c$ by $b + c$ and getting $1 + \frac{1}{2}x$ instead of $1 + x$: remember x already contains the factor 2.
- Using $xy + yz + zx < 4$ (an *upper* bound) to try to bound S from below; the inequalities have to point the other way, as above.

Beyond STEP

The fact that a square matrix kills a non-zero vector exactly when its determinant is zero is the heart of linear algebra: it is how eigenvalues are found, via $\det(A - \lambda I) = 0$. The bound $S \geq 3$ in part (ii) is Nesbitt's inequality, $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$, in disguise. For the linear algebra, see Strang, *Introduction to Linear Algebra*; for more inequalities of this kind, Siklos, *Advanced Problems in Mathematics*, is excellent practice.

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